Spectroscopy

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1 Introduction

Spectroscopy deals with the analysis of materials or gases by measurements of their absorption, reflection, or emission spectra. In this experiment, a grating spectrometer apparatus—consisting of a light source, collimator, grating, and telescope—is used to study the emission spectra of low-pressure gases in which atoms are excited by an electrical discharge.

The primary purpose of this experiment is to explore these atomic and molecular emissions. Firstly, the grating spectrometer is calibrated using the well-known emission doublet of sodium (Na), which has two closely spaced lines at 5889.95 Å and 5895.92 Å. The fundamental principle governing this is the grating equation

$$m\lambda = d(\sin\theta) \tag{1}$$

where d represents the grating constant, m is the order of diffraction, and θ is the diffraction angle for a specific wavelength λ. Next, utilizing the calibrated spectrometer, the emission lines of a hydrogen source are measured, alongside those of helium, nitrogen, and an unknown gas. These measurements are then compared against established spectral data, and the unknown gas is identified based on its emission characteristics. Lastly, the hydrogen spectrum is analyzed to determine the Rydberg constant, R. This is calculated using the quantized energies of an electron in a Coulomb potential, given by

$$\frac{1}{\lambda} = R\left(\frac{1}{4} - \frac{1}{n^2}\right) \tag{2}$$

where n is the principal quantum number of the Balmer series transitions for each observed spectral line.

2 Methods

Experimental Setup

The experimental setup consists of a collimator, a diffraction grating (600 lines/mm), and a telescope mounted on a rotating stage. A discharge tube is placed at the focal point of the collimator to serve as the light source. The light passes through the collimator to create a parallel beam, strikes the diffraction grating, and is then observed through the rotating telescope, allowing for the measurement of the angles corresponding to different spectral lines.

Data Collection

The spectrometer is first calibrated using a sodium lamp, where a reference initial angle is recorded with the telescope approximately normal to the grating. The angles for the first-order diffraction lines are measured, and the corresponding wavelengths are calculated using the grating equation. The calculated wavelengths are then compared to the known values to determine whether the starting angle is accurate or if an adjustment is necessary.

After calibration, the sodium lamp is replaced with discharge tubes containing hydrogen, helium, nitrogen, and an unknown gas (referred to as Unknown E). For each gas, the telescope is swept across the viewing field to locate the visible emission lines, and the corresponding angles are recorded.

Data Analysis

The raw angular data is processed using a Python script 1 that automates the calibration, wavelength calculation, and determination of the Rydberg constant for hydrogen.

The script first calculates the theoretical diffraction angle for the sodium reference, θ_{ref}, using Equation 1 rearranged for a first-order ($m = 1$) diffraction:

$$\theta_{\text{ref}} = \arcsin\left(\frac{\lambda_{\text{Na}}}{d}\right)$$

where d is the nominal grating constant (calculated from the 600 lines/mm specification) and λ_{Na} is the average wavelength of the sodium. The wavelength λ_{Na} is taken as an average since the two lines are very close together, and the primitive spectrometer's resolution may not distinguish them clearly. The true zero angle of the spectrometer is then determined by offsetting the

experimentally measured sodium angle by θ_{ref}, depending on which side of the central maximum the measurement was taken.

For every subsequent emission line observed in the hydrogen, helium, nitrogen, and unknown gas, the absolute angle of diffraction, θ, is found by taking the absolute difference between the measured angle and the calculated zero angle. The emission wavelength, λ, is then computed using the calibrated angle:

$$\lambda = d \sin(\theta)$$

The hydrogen spectrum is isolated to calculate the experimental Rydberg constant, R. The script associates the observed colors of the Balmer series with their corresponding principal quantum numbers: Red ($n = 3$), Cyan ($n = 4$), Blue ($n = 5$), and Violet ($n = 6$). Using Equation 2, a linear regression is performed, plotting $\frac{1}{\lambda}$ against $\left(\frac{1}{4} - \frac{1}{n^2}\right)$ to extract the value of R from the slope. Finally, the script calculates the percent error by comparing this experimental slope to the theoretical Rydberg constant, $R_{\text{theoretical}} = 1.097\,373 \times 10^7\,\text{m}^{-1}$ and generates a regression plot of the results.

Listing 1: Python script for analyzing emission spectra and determining experimental Rydberg constants

```python
import json
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path
from typing import Any, Sequence
from scipy.stats import linregress
from uncertainties import ufloat
from uncertainties.umath import asin, degrees, radians, sin
import uncertainties.unumpy as unp


def load_data(filepath: Path | str) -> dict[str, Any]:
    """Load spectroscopy observations from a JSON file.

    :param filepath: Path to the JSON data file.
    :returns: Parsed JSON content.
    """
    with open(filepath, 'r') as f:
        return json.load(f)


def calibrate_zero_angle(
        d_nm: float,
        ref_wavelength_nm: float,
        measured_angle_deg: Any,
```

```python
        start_angle_deg: Any,
        order: int = 1,
) -> Any:
    """
    Calculates the true zero angle of the rotating stage using a reference wavelength.

    :param d_nm: The grating constant in nanometers.
    :param ref_wavelength_nm: Reference wavelength used for calibration in
        nanometers.
    :param measured_angle_deg: Measured angle for the reference wavelength in
        degrees.
    :param start_angle_deg: Initial recorded stage angle in degrees.
    :param order: Diffraction order (default = 1).
    :returns: Calibrated zero-angle position in degrees.
    """
    # Calculate theoretical diffraction angle for the reference wavelength
    theta_ref_rad = asin((order * ref_wavelength_nm) / d_nm)
    theta_ref_deg = degrees(theta_ref_rad)

    # Determine true zero angle based on which side of the central max it was measured
    if measured_angle_deg > start_angle_deg:
        true_zero_angle = measured_angle_deg - theta_ref_deg
    else:
        true_zero_angle = measured_angle_deg + theta_ref_deg

    return true_zero_angle


def calculate_wavelength(
    d_nm: float,
    angle_degrees: Any,
    true_zero_angle_degrees: Any,
    order: int = 1,
) -> Any:
    """
    Calculates wavelength using the calibrated zero angle.

    :param d_nm: The grating constant in nanometers.
    :param angle_degrees: Measured diffraction angle in degrees.
    :param true_zero_angle_degrees: Calibrated zero-angle position in degrees.
    :param order: Diffraction order (default = 1).
```

```python
    :returns: Calculated wavelength in nanometers.
    """
    theta_rad = radians(abs(angle_degrees - true_zero_angle_degrees))
    return (d_nm * sin(theta_rad)) / order


def analyze_rydberg_constant(
    x_vals: Sequence[Any],
    y_vals: Sequence[Any],
    out_path: Path,
) -> None:
    """Fit the Balmer relation and save a regression plot.

    :param x_vals: Balmer-series term values ``(1/2^2 - 1/n^2)``.
    :param y_vals: Inverse wavelengths in ``nm^-1``.
    :param out_path: Output directory for generated figures.
    :returns: None.
    """
    # Extract nominal values for plotting/regression
    x_nom = unp.nominal_values(x_vals)
    y_nom = unp.nominal_values(y_vals)
    y_std = unp.std_devs(y_vals)

    # Linear regression: 1/lambda = R * (1/4 - 1/n^2)
    if any(y_std > 0):
        weights = 1 / y_std
        idx = np.isfinite(weights)
        slope, intercept = np.polyfit(np.array(x_nom)[idx], np.array(y_nom)[idx], 1, w=weights[idx])
        cov = np.polyfit(np.array(x_nom)[idx], np.array(y_nom)[idx], 1, w=weights[idx], cov=True)[1]
        slope_err = np.sqrt(cov[0, 0])
        slope_u = ufloat(slope, slope_err)
    else:
        slope, intercept, r_value, p_value, std_err = linregress(x_nom, y_nom)
        slope_u = ufloat(slope, std_err)

    R_experimental = slope_u * 1e9
    R_theory = 1.097373e7  # m^-1

    print(f"\nExperimental Rydberg Constant R: {R_experimental} m^-1")
    print(f"Theoretical Rydberg Constant R: {R_theory:.4e} m^-1")

    # Plot linear regression
```

```python
    plt.figure(figsize=(8, 6))
    if any(y_std > 0):
        plt.errorbar(x_nom, y_nom, yerr=y_std, fmt='o', color='red', zorder=5, label='Experimental Data (with uncertainty)')
    else:
        plt.scatter(x_nom, y_nom, color='red', zorder=5, label='Experimental Data')

    # Calculate line of best fit
    fit_x = np.linspace(0, max(x_nom) * 1.1, 100)
    fit_y = slope * fit_x + intercept

    plt.plot(fit_x, fit_y, color='blue', linestyle='--', label=f'Linear Fit\n$y = {slope:.4e}x + {intercept:.4e}$')

    plt.title('Hydrogen Balmer Series - Rydberg Constant Regression')
    plt.xlabel(r'$\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$')
    plt.ylabel(r'$\frac{1}{\lambda}$ (nm$^{-1}$)')
    plt.grid(True, linestyle=':', alpha=0.7)
    plt.legend()
    plt.tight_layout()

    plot_path = out_path / "rydberg_regression.png"
    plt.savefig(plot_path)
    plt.close()
    print(f"Saved regression plot to {plot_path}")


def perform_calibration(observations: dict[str, Any], d_nominal_nm: float) -> Any:
    """Compute stage zero-angle from the sodium reference line.

    :param observations: Observation data grouped by gas name.
    :param d_nominal_nm: Nominal grating spacing in nanometers.
    :returns: Calibrated zero angle in degrees.
    """
    print(f"--- Calibration ---")
    na_data = observations['Sodium']
    na_start = ufloat(na_data['starting_angle'], 1/6)
    na_angle = ufloat(na_data['lines'][0]['angle'], 1/6)

    # Average wavelength of the Sodium doublet
    na_avg_wl_nm = (588.995 + 589.592) / 2.0

    # Calculate the true zero angle
    true_zero_angle = calibrate_zero_angle(d_nominal_nm, na_avg_wl_nm, na_angle, na_start)
```

```python
    angle_offset = abs(true_zero_angle - na_start)

    print(f"Nominal d: {d_nominal_nm:.2f} nm")
    print(f"Recorded Start Angle: {na_start} deg")
    print(f"True Zero Angle: {true_zero_angle} deg")
    print(f"Zero Angle Error: {angle_offset} deg")

    return true_zero_angle


def analyze_hydrogen(
    observations: dict[str, Any],
    d_nominal_nm: float,
    true_zero_angle: Any,
    out_path: Path,
) -> None:
    """Analyze hydrogen Balmer lines and estimate the Rydberg constant.

    :param observations: Observation data grouped by gas name.
    :param d_nominal_nm: Nominal grating spacing in nanometers.
    :param true_zero_angle: Calibrated zero-angle position in degrees.
    :param out_path: Output directory for generated plots.
    :returns: None.
    """
    print(f"\n--- Hydrogen Analysis (Rydberg Constant) ---")
    h_data = observations['Hydrogen']

    color_to_n = {
        "Red": 3,
        "Cyan": 4,
        "Blue": 5,
        # NOTE: Omit violet due to anomalous results
        # "Violet": 6
    }

    x_vals = []
    y_vals = []

    for line in h_data['lines']:
        color = line['color']
        measured_angle = ufloat(line['angle'], 1/6)
        n = color_to_n.get(color)
        if n:
            wl = calculate_wavelength(d_nominal_nm, measured_angle, true_zero_angle)
            true_angle = abs(measured_angle - true_zero_angle)
            inv_wl = 1.0 / wl
```

```python
            term = (0.25 - 1.0 / (n ** 2))
            x_vals.append(term)
            y_vals.append(inv_wl)
            print(f"Color: {color:7} | n={n} | Measured Angle={measured_angle} deg | True Angle={true_angle} deg | Wavelength={wl} nm")

    analyze_rydberg_constant(x_vals, y_vals, out_path)


def analyze_helium(
    observations: dict[str, Any],
    d_nominal_nm: float,
    true_zero_angle: Any,
) -> None:
    """Compute wavelengths for measured helium emission lines.

    :param observations: Observation data grouped by gas name.
    :param d_nominal_nm: Nominal grating spacing in nanometers.
    :param true_zero_angle: Calibrated zero-angle position in degrees.
    :returns: None.
    """
    print(f"\n--- Helium Analysis ---")
    he_data = observations['Helium']

    for line in he_data['lines']:
        color = line['color']
        measured_angle = ufloat(line['angle'], 1/6)

        wl = calculate_wavelength(d_nominal_nm, measured_angle, true_zero_angle)
        true_angle = abs(measured_angle - true_zero_angle)
        print(f"Color: {color:7} | Measured Angle={measured_angle} deg | True Angle={true_angle} deg | Wavelength={wl} nm")


def analyze_unknown_e(
    observations: dict[str, Any],
    d_nominal_nm: float,
    true_zero_angle: Any,
) -> None:
    """Compute wavelengths for measured lines of the unknown gas sample.

    :param observations: Observation data grouped by gas name.
    :param d_nominal_nm: Nominal grating spacing in nanometers.
    :param true_zero_angle: Calibrated zero-angle position in
```

```python
        degrees.
    :returns: None.
    """
    print(f"\n--- Unknown_E Analysis ---")
    unk_data = observations['Unknown_E']

    for line in unk_data['lines']:
        color = line['color']
        measured_angle = ufloat(line['angle'], 1/6)

        wl = calculate_wavelength(d_nominal_nm, measured_angle, true_zero_angle)
        true_angle = abs(measured_angle - true_zero_angle)
        print(f"Color: {color:10} | Measured Angle={measured_angle} deg | True Angle={true_angle} deg | Wavelength={wl} nm")


def main() -> None:
    """Run calibration and spectroscopy analysis pipeline.

    :returns: None.
    """
    data_path = Path(__file__).parent.parent / "data" / "spectrum.json"
    out_path = Path(__file__).parent.parent / "out"
    out_path.mkdir(exist_ok=True, parents=True)

    data = load_data(data_path)

    params = data['experimental_parameters']
    observations = data['observations']

    # Get calibration
    d_nominal_nm = 1e6 / params['diffraction_grating_lines_per_mm']
    true_zero_angle = perform_calibration(observations, d_nominal_nm)

    # Analyze gasses
    analyze_hydrogen(observations, d_nominal_nm, true_zero_angle, out_path)
    analyze_helium(observations, d_nominal_nm, true_zero_angle)
    analyze_unknown_e(observations, d_nominal_nm, true_zero_angle)


if __name__ == "__main__":
    main()
```

3 Raw Data

The raw data collected from the spectrometer for each gas is summarized in Table 1. Each entry includes the gas type, the base angle at which the gas was observed, the color of the spectral line, and the corresponding observation angle.

Table 1: Spectral Emission Observations for Various Gases

Gas	**Base Angle (°)**	**Line Color**	**Observation Angle (°)**
Sodium	295.0	Doublet (Na)	313.6
Hydrogen	295.0	Violet	293.0
		Blue	277.5
		Cyan	276.6
		Red	269.0
Helium	295.0	Purple	277.0
		Cyan	276.0
		Green	275.3
		Yellow	275.0
		Orange	271.6
		Red	268.6
Nitrogen	295.0	Distinct	293.0
		Blue lines (max)	278.6
		Blue lines (min)	276.0
Unknown E	295.0	Blue 1 & 2	277.6
		Blue 3	277.3
		Blue 4 & 5	277.0
		Green	273.0
		Orange	271.6
		Red	269.0

4 Analysis and Results

The calibration of the spectrometer using the sodium doublet yielded a true zero angle of $(292.95 \pm 0.17)^\circ$, compared to the recorded starting angle of $(295.00 \pm 0.17)^\circ$. This resulted in a zero-angle error of $(2.05 \pm 0.24)^\circ$. The nominal grating constant d, was calculated to be 1667.67 nm. This

calibrated zero angle was applied to all subsequent measurements to find the true diffraction angles for the observed spectral lines.

The observed emission lines for the hydrogen Balmer series are detailed in Table 2. Applying a linear regression to this data, the experimental Rydberg constant was determined to be $R_{\text{experimental}} = (1.17 \pm 0.14) \times 10^7 \text{ m}^{-1}$. The linear fit to extract this experimental constant is plotted in Figure 1.

Table 2: Measured angles and calculated wavelengths for the Hydrogen emission spectrum.

Color	n	Measured Angle (°)	True Angle (°)	Wavelength (nm)
Violet	6	293.00 ± 0.17	0.05 ± 0.24	1 ± 7
Blue	5	277.50 ± 0.17	15.45 ± 0.24	444 ± 7
Cyan	4	276.60 ± 0.17	16.35 ± 0.24	469 ± 7
Red	3	269.00 ± 0.17	23.95 ± 0.24	677 ± 6

Following the hydrogen analysis, the emission spectra for helium and the unknown gas were measured. The respective measured angles corrected true diffraction angles, and computed wavelengths for these gases are presented in Table 3 and Table 4. The emission line for nitrogen were observed but not precisely measured due to their diffuse nature, so they are not included in the tables.

Table 3: Measured angles and calculated wavelengths for the Helium emission spectrum.

Color	Measured Angle (°)	True Angle (°)	Wavelength (nm)
Purple	277.00 ± 0.17	15.95 ± 0.24	458 ± 7
Cyan	276.00 ± 0.17	16.95 ± 0.24	486 ± 7
Green	275.30 ± 0.17	17.65 ± 0.24	505 ± 7
Yellow	275.00 ± 0.17	17.95 ± 0.24	514 ± 7
Orange	271.60 ± 0.17	21.35 ± 0.24	607 ± 6
Red	268.60 ± 0.17	24.35 ± 0.24	687 ± 6

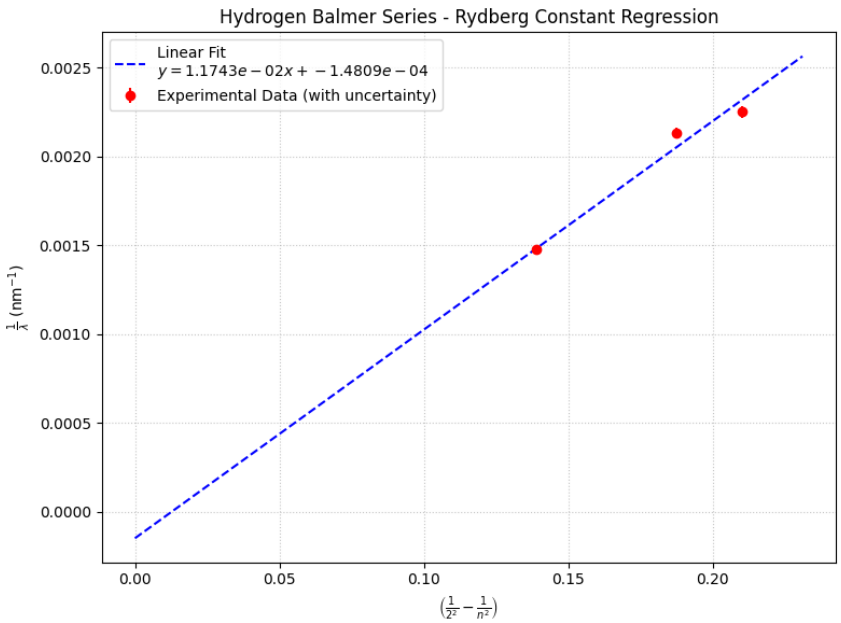

Figure 1: Linear regression to determine the experimental Rydberg constant from the Hydrogen emission lines.

Table 4: Measured angles and calculated wavelengths for Unknown Gas E.

Color	Measured Angle (°)	True Angle (°)	Wavelength (nm)
Blue 1 & 2	277.60 ± 0.17	15.35 ± 0.24	441 ± 7
Blue 3	277.30 ± 0.17	15.65 ± 0.24	450 ± 7
Blue 4 & 5	277.00 ± 0.17	15.95 ± 0.24	458 ± 7
Green	273.00 ± 0.17	19.95 ± 0.24	569 ± 6
Orange	271.60 ± 0.17	21.35 ± 0.24	607 ± 6
Red	269.00 ± 0.17	23.95 ± 0.24	677 ± 6

5 Discussion

The primary objective of this experiment was to empirically determine the Rydberg constant through the spectral analysis of hydrogen. By plotting the inverse of the calculated wavelengths against the $\left(\frac{1}{4} - \frac{1}{n^2}\right)$ term from the Rydberg formula, a clear linear relationship was established. The slope of this regression yielded an experimental value of $R_{\text{experimental}} = (1.17 \pm 0.14) \times 10^7\ \text{m}^{-1}$. When evaluated against the accepted NIST value of $1.097\,373 \times 10^7\,\text{m}^{-1}$ [1], the experimental result carries a percent error of approximately 6.6%. Because the theoretical value falls comfortably within the experimental uncertainty bounds, the methodology successfully validates the quantized energy level model of the hydrogen atom, despite the presence of underlying systematic errors in the specific wavelength calculations.

An analysis of the raw wavelength data reveals a consistent systematic shift. The calculated wavelengths for the hydrogen Balmer series — (677 ± 6) nm (Red), (469 ± 7) nm (Cyan), and (444 ± 7) nm (Blue) — are uniformly higher than the accepted values of 656.28 nm (H_α), 486.13 nm (H_β), and 434.05 nm (H_γ), respectively [2]. While these three lines adhered strongly to the linear quantum model and were used to calculate the Rydberg constant, the measurement for the violet line (H_δ) was explicitly excluded from the final regression. Its calculated wavelength exhibited an extreme deviation from the expected value. Because the other three lines possessed a consistent offset rather than a massive divergence, this anomaly is highly unlikely to stem from baseline systematic errors. Instead, it almost certainly arose a discrete error in the angular measurement itself.

The consistent overestimation seen in the red, cyan, and blue lines strongly suggests an inaccuracy in the grating constant, d, which was atken as exactly 1667.67 nm based on the 600 lines/mm specification. Any manufacturing tolerance in the grating would systematically scale every subsequent calculation. This systemic shift could be worsened by angular misalignment; if the plane of the diffraction grating was not perfectly orthogonal to the incident collimated beam, the effective grating spacing would be altered, introducing a cosine-dependent error across all measurements.

Beyond systematic biases, the precision of the results was constrained by instrumental errors. The mechanical resolution of the spectrometer's angular scale dictated a baseline uncertainty of $\pm 0.17°$ for every reading. This error was compounded by the physical tuning of the collimator slit. While a wider slit is necessary to capture enough luminosity to view the fainter higher-order lines, it inherently broadens the angular width of the spectral peaks, making it difficult to align the viewing telescope exactly on the cen-

ter of maximum intensity. Finally, the mathematical analysis relied on the vacuum approximation of the grating equation, omitting the refractive index of the ambient air in the laboratory. While this omission only shifts the expected wavelengths by a small fraction, it represents a fundamental limitation in the accuracy of the employed mathematical model.

The analysis of the helium spectrum yielded much less precise results overall. While a few of the calculated wavelengths fell reasonably close to expected values, the majority of the measured helium lines deviated significantly from standard NIS tables [3]. This suggests that the baseline systematic errors were compounded by the difficulty of resolving helium's denser array of emission lines, potentially leading to the misidentification of adjacent spectral peaks or higher-order diffractions.

Despite the broader inaccuracies encountered with the helium data, the emission profile for the unknown gas (E) provided enough relative structural data to successfully determine its identity. By comparing the specific sequence of the calculated wavelengths — particularly the dense cluster of multiple blue lines alongside the distinct green, orange, and red emissions — to standard reference spectra [4], the sample was confidently identified as krypton.

Beyond the quantitative measurements of these distinct gases, a qualitative observation of nitrogen gas highlighted the difference between atomic and molecular emissions. Unlike the sharp, discrete peaks characteristic of the monatomic samples, the nitrogen emission appeared as a broad, continuous band. This dense clustering arises directly from its molecular structure. As a diatomic molecule, nitrogen possesses additional degrees of freedom allowing for rotational and vibrational internal motion. The quantum mechanical transitions between these numerous rotational-vibrational states produce a spectrum with so many closely spaced lines that it visually blurs into a continuous band.

6 Conclusion

This experiment successfully utilized a calibrated grating spectrometer to analyze atomic emission spectra and validate the quantum mechanical model of the hydrogen atom. By measuring the diffraction angles of the hydrogen Balmer series and applying a linear regression to the Rydberg formula, the experimental Rydberg constant was determined to be $R_{\text{experimental}} = (1.17 \pm 0.14) \times 10^7$. This result carries a 6.6% error compared to the accepted NIST standard, confirming the theoretical model while highlight-

ing a consistent systematic overestimation of wavelengths. This systematic shift was primarily attributed to inaccuracies in the nominal grating constant and potential orthogonal misalignment of the spectrometer. Despite these instrumental limitations, the spectral analysis methodology proved robust enough to conclusively identify an unknown gas sample as krypton based on its unique atomic fingerprint. Furthermore, qualitative observations of nitrogen successfully demonstrated the physical distinction between the sharp, discrete energy transitions of monatomic gases and the continuous rotational-vibrational emission bands characteristic of diatomic molecules.

References

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[4] National Institute of Standards and Technology. "Strong lines of krypton (kr)." (2025), [Online]. Available: https://physics.nist.gov/PhysRefData/Handbook/Tables/kryptontable2_a.htm.